



**UNIVERSITÉ
DE GENÈVE**

GENEVA SCHOOL OF ECONOMICS
AND MANAGEMENT

Introduction to Macroeconomics

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Problem Set 3

2025

Exercise 1

In a closed economy, we know that the GDP is equal to 400, consumption is equal to 250, government expenditure equals 100 and private saving is 130. Which of the following statements is correct ?

- (a) National saving equals 210, public saving equals 80 and taxes equal 20
- (b) Investment equals 130, budget deficit equals 80 and taxes 50.
- (c) Budget deficit equals 80, investment equals 50 and taxes equal 20.
- (d) We do not have enough data to determine the macroeconomic equilibrium

Exercise 2

Consider the following events :

1. Future income becomes more uncertain, which leads the households to start saving more
 2. An investment tax credit, which is a tax advantage given to any firm that decides to invest, disappears.
 3. The government, which previously had a balanced budget, now runs a budget surplus.
- (a) What would be the effect of each of these events on the supply and demand of loanable funds and on the interest rate ?
 - (b) Suppose now that following the disappearance of the investment tax credit, the government ends up running a surplus budget (a combination of events 2. and 3.). What can you claim for sure concerning the level of interest rate and the quantity of loanable funds loaned at the equilibrium in this case ?

Exercise 3

What is the economic phenomenon called 'crowding out'?

- (a) When the number of mutual funds increases, the economic activity by private banks as financial intermediaries falls.
- (b) When the government increases taxes on investment, which will lead to a decline in private investment.
- (c) When the government borrows to finance its budget deficit, which reduces the amount of loanable funds available to finance private investment. As a result, there is a decline in private investment.
- (d) None of the above.

Exercise 4

The money multiplier allows the Central Bank to evaluate the impact of any given increase of the reserves (here called M_0) on the money supply (here called M_1). A common assumption is that the households decide not to keep any liquidity and deposit all the currency they hold in the commercial banks, but here it is not the case (they decide to hold a 40% of the currency in cash). If the reserve requirement is $r = 10\%$ and the share of deposits held by households is $c = 40\%$, which of the following propositions is true?

- (a) The money multiplier is between 2 and 2.2.
- (b) If $M_0 = 400\$$ then $M_1 = 4000\$$.
- (c) The money multiplier is always higher than in the case where $c = 0$.
- (d) None of the above.

Exercise 5

Which combination of the central bank's measures will necessarily increase money supply?

- (a) buying bonds and decreasing the reserve requirement
- (b) buying bonds and increasing the interest rate
- (c) selling bonds and decreasing the interest rate
- (d) increasing the reserve requirement and decreasing the interest rate

Exercise 6

According to the efficient market hypothesis, what is the probable effect of each of the following events on the given company's stock market share price?

- (a) Several of the companies operating in the same sector as Company A announce that their sales have been higher than predicted.
- (b) Change in the fiscal legislation which was adopted last year reduces Company B's profit this year.
- (c) Company C announces that it is confident to achieve the profits that were initially planned.
- (d) A legal dispute, whose outcome was uncertain, is resolved in the favour of Company D.

Additional exercises - Not covered in seminar

Exercise 7

Which of the following central bank policies leads to an increase in the money supply?

- (a) Increasing the reserve requirements.
- (b) Selling government bonds.
- (c) Raising the refinancing rate.
- (d) Purchasing foreign currency in international markets.

Exercise 8

In a closed economy, we know that the GDP is equal to 300, consumption is equal to 180, investment is equal to 50 and taxes are equal to 30. Which of the following statements is correct?

- (a) The government purchases are equal to 30, the saving is equal to 60, of which 20 is private saving and 40 is public saving.
- (b) The government runs a budget deficit of 20 and its purchases are equal to 30.
- (c) The private saving is equal to 40 and the budget surplus of the government is equal to 10.
- (d) None of the previous statements is correct.

Exercise 9

The equilibrium interest rate is currently 4%. Which of the following changes in the loanable funds market would lead to a higher interest rate for sure?

- (a) A decrease in the demand for loanable funds.
- (b) An increase in the demand for loanable funds and a decrease in the supply of loanable funds.
- (c) A decrease in the demand and in the supply for loanable funds.
- (d) An increase in the demand and the supply for loanable funds.